Abdelaziz Omar

Software Developer

Faisal, Giza, Egypt • abdelaziz.omar405@gmail.com • +201150842586 • abdelaziz-omar.com

PROFESSIONAL EXPERIENCE

Software Engineer ***September 2025 - Present***

VertixSoft | **Full-Time | On Sit** ***Nasr City, Cairo, Egypt***

- Developing and maintaining backend services and RESTful APIs to support business requirements across multiple product lines.
- Collaborating with cross-functional teams to design scalable system architecture and ensure code quality through continuous improvement.

Backend Developer ***July 2025 - August 2025***

Agricultural Bank of Egypt | **Internship | On Sit** ***Dokki, Giza, Egypt***

- Attended training sessions covering backend development best practices and secure API design principles applied within financial systems.
- Gained exposure to cloud-based deployment and system integration workflows in a professional banking environment.

Backend Developer ***August* 2024 - *September* 2024**

Robusta Studio | **Internship | Hybrid** ***New Cairo, Cairo, Egypt***

- Completed an intensive backend development track covering RESTful API design, database management, and authentication best practices.
- Built and delivered a backend project as part of the program's final assessment, applying industry-standard Node.js development workflows.

PROJECTS

URL Shortener & Analytics API

- Built a URL shortening service with analytics tracking for visit count, referrer, and timestamps, backed by PostgreSQL and reporting endpoints for per-link insights.
- Integrated Redis for redirect caching to reduce database load on high-traffic links, and implemented Redis-based rate limiting to protect the shortening endpoint from abuse.
- Secured REST endpoints with JWT authentication, input validation, proper HTTP status codes, and consistent error response handling.
- **Technologies: Node.js, Express.js, PostgreSQL, Redis, JWT.**

Real-Time Chat API

- Engineered a real-time messaging backend supporting multiple chat rooms and concurrent users via WebSockets, with persistent message history stored in PostgreSQL.
- Implemented JWT authentication for both HTTP and WebSocket connections, middleware-based request validation, and role-based access control for room creation and management.
- Designed a normalized PostgreSQL schema for users, rooms, memberships, and messages with efficient querying for pagination and room-based message retrieval.
- **Technologies: Node.js, Express.js, WebSockets, PostgreSQL, JWT.**

E-Commerce REST API

- Built a secure e-commerce backend supporting product catalog, shopping cart, order management, and user account workflows with full RESTful CRUD operations.
- Implemented JWT-based authentication and role-based authorization (customer / admin), password hashing with Bcrypt.js, and centralized error handling with structured API responses.
- Added pagination, filtering, sorting, and order-related email notifications using Nodemailer, while following backend security best practices.
- **Technologies: Node.js, Express.js, MongoDB, JWT, Bcrypt.js, Nodemailer.**

EDUCATION

Culture & Science City, Faculty of Computer Science, ***Giza, Egypt***

B.Sc. IN COMPUTER SCIENCE AND INFORMATION ***September2022 – June 2026 (Expected)***

- Anticipated GPA: 3.08 / 4.0

SKILLS & OTHER

- **Programming Languages:** JavaScript, TypeScript, Python, C#, C/C++.
- **Frameworks**: Node.js, Express.js, REST APIs, JWT.
- **Databases:** SQL, MySQL, MongoDB, PostgreSQL.
- **Volunteering:** Instructor at Trainopia | 2023 – 2025. Delivered technical workshops to new participants.

COLLEGE PROJECTS

Real-Time Object Detection and Tracking System

3rd year college project | Image Processing Course | Mar 2025

- Built a real-time detection system using edge detection, contour analysis, and background subtraction to track objects across video frames under varying lighting.
- **Tools and Technologies: Python, OpenCV, NumPy.**

Process Scheduling Simulator

3rd year college project | Operating Systems Course | Jan 2025

- Simulated CPU scheduling algorithms (FCFS, SJF, Round Robin, Priority) with Gantt chart output and metrics including average waiting time and CPU utilization.
- **Tools and Technologies: C, Linux, POSIX Threads.**

Smart Campus Navigation App

2nd year college project | Human-Computer Interaction Course | Apr 2024

- Designed and prototyped a mobile navigation app applying HCI principles, with user research and usability testing sessions to iteratively improve the experience.
- **Tools and Technologies: Figma, Adobe XD, Usability Testing Methodologies.**

Pathfinding Visualizer

2nd year college project | Data Structures and Algorithms Course | Apr 2024

- Built an interactive grid visualizer for shortest-path algorithms (Dijkstra's, BFS, A*) implementing priority queues and adjacency lists from scratch.
- **Tools and Technologies: C++, Data Structures, Algorithms, SFML.**

University Library Management System

2nd year college project | Database Management Course | Jan 2024

- Designed a normalized relational database schema (3NF) with stored procedures and triggers to automate fine calculation and book availability tracking.
- **Tools and Technologies: SQL Server, MySQL, Entity-Relationship Diagrams.**

Hospital Appointment Booking System

1st year college project | Software Engineering Course | Apr 2023

- Developed a system following the full SDLC with requirements gathering, UML design, implementation, and testing for patient registration and doctor scheduling.
- **Tools and Technologies: C++, UML, Microsoft Visio.**

Student Course Management System

1st year college project | Programming Techniques Course | Mar 2023

- Built a console-based system to manage students, courses, and grades applying OOP principles including encapsulation, inheritance, and polymorphism.
- **Tools and Technologies: C++, Object-Oriented Programming, Visual Studio 2019.**

References available upon request